

Exercise 01: Sensors

Task 1.1: Piezoelectric accelerometer under shock + drift

You're designing a 10 g full-scale accelerometer for impact testing using a PZT stack in the longitudinal mode (use $d_{33} = 593 \text{ pC/N}$). The proof mass is $m = 5 \text{ g}$. The shock is a half-sine $a(t) = a_{\text{pk}} \sin(\pi t/T)$ with $a_{\text{pk}} = 500 \text{ m/s}^2$, $T = 4 \text{ ms}$. The ceramic disk has active area $A = 100 \text{ mm}^2$, thickness $t = 0.5 \text{ mm}$, relative permittivity $\varepsilon_r = 1500$. Assume quasi-static force transfer $F(t) = m a(t)$ and a charge amplifier with $C_f = 100 \text{ pF}$, ideal op-amp.

1.1.1 Compute the sensor charge $Q(t)$ and amplifier output $v_o(t)$.

Solution:

Force:

$$F(t) = m a(t) = m a_{\text{pk}} \sin\left(\frac{\pi t}{T}\right).$$

Piezoelectric charge (direct effect, quasi-static transfer):

$$Q(t) = d_{33} F(t) = d_{33} m a_{\text{pk}} \sin\left(\frac{\pi t}{T}\right).$$

Charge amplifier ideal output:

$$v_o(t) = -\frac{Q(t)}{C_f} = -\frac{d_{33} m a_{\text{pk}}}{C_f} \sin\left(\frac{\pi t}{T}\right).$$

Calculating using the given values:

$$F_{\text{peak}} = m a_{\text{pk}} = 0.005 \times 500 = 2.5 \text{ N},$$

$$Q_{\text{peak}} = d_{33} F_{\text{peak}} = 593 \times 10^{-12} \times 2.5 = 1.4825 \times 10^{-9} \text{ C} = 1482.5 \text{ pC},$$

$$v_{o,\text{peak}} = -\frac{Q_{\text{peak}}}{C_f} = -\frac{1.4825 \times 10^{-9}}{100 \times 10^{-12}} = -14.825 \text{ V}.$$

Therefore, the time-domain results are:

$$Q(t) = 1.4825 \times 10^{-9} \sin\left(\frac{\pi t}{4 \text{ ms}}\right) \text{ C},$$

$$v_o(t) = -14.825 \sin\left(\frac{\pi t}{4 \text{ ms}}\right) \text{ V},$$

for $0 \leq t \leq 4 \text{ ms}$ (and zero outside the half-sine).

Note: The output $v_o(t)$ follows a negative half-sine waveform with a peak amplitude of approximately -14.825 V , due to the inverting nature of the charge amplifier.

1.1.2 Estimate the sensor capacitance C_s and discuss how C_s versus C_f affects the scale factor and low-frequency droop.

Solution:

Ideal charge amplifier: For an ideal op-amp, the output depends only on the generated charge and the feedback capacitor:

$$v_o(t) = -\frac{Q(t)}{C_f}$$

so ideally the sensor capacitance C_s does not affect the scale factor, which is set by $1/C_f$.

$$C_s = \frac{\varepsilon_0 \varepsilon_r A}{t} \approx 2.7 \text{ nF}, \quad \text{with } C_s \gg C_f, \text{ the op-amp forces the signal charge onto } C_f : \quad |v_o| = \frac{Q}{C_f}.$$

Smaller C_f increases sensitivity but also raises noise and saturation risk.

1.1.3 If the device warms by $+40^\circ\text{C}$, qualitatively discuss the expected sensitivity shift and explain why a constant force cannot be measured indefinitely with a piezoelectric sensor.

Solution:

Event ID	OS ID	Hazard	Potential Effect
HE1	OS2	H1	Sudden tilt, loss of stability $\rightarrow$ possible loss of control
HE2	OS3	H2	Incorrect leveling $\rightarrow$ reduced ground clearance, bottoming out
HE3	OS2	H3	Continuous height oscillation $\rightarrow$ handling instability
HE4	OS4	H4	Unintended height adjustment during braking $\rightarrow$ change in braking dynamics
HE5	OS1	H5	Vehicle collapses on suspension $\rightarrow$ risk of injury when entering/exiting

1.1.4 Perform a Risk Assessment based on the Severity (S), Exposure (E) and Controllability (C) using ISO 26262 Framework. Also derive ASIL (Automotive Safety Integrity Level).

Solution:

ISO 26262 is the international standard for functional safety of electrical and electronic (E/E) systems in production automobiles.

It provides a framework and set of processes to ensure that automotive systems are designed, developed, and validated so that they do not cause unreasonable risk to human life, even if hardware or software faults occur.

It ensures safety by design.

The standard is particularly important for systems like Advanced Driver Assistance Systems (ADAS) and autonomous driving technologies.

ASIL (Automotive Safety Integrity Level)

A measure of required risk reduction — how “safe” a function must be.

Event ID	OS ID	Hazard
A	Low	Seat heating malfunction
B	Moderate	Wrong tire pressure reading
C	High	Loss of steering assist
D	Very High	Unintended acceleration
QM	Quality Management	Non-safety-critical (e.g., radio)

Level	Description	Typical consequence	Example
S1	Light to moderate injuries	Minor discomfort, property damage	Slight height deviation at low speed
S2	Severe injuries (survivable)	Broken bones, major bruising	Sudden vehicle tilt at low speed
S3	Life-threatening or fatal injuries	Could lead to death or serious injury	Loss of vehicle control at high speed

Table 1.1.1: S - Potential extent of harm to a person if the hazardous event takes place.

Level	Description	Frequency of Situation	Example
E1	Very low	Happens rarely	Off-road mode used only occasionally
E2	Medium	Occurs sometimes	City driving (occasional highway)
E3	High	Happens frequently	Normal driving on road (daily)
E4	Very High	Almost always	Engine running, normal driving

Table 1.1.2: E - Probability of a vehicle being in a particular operational situation.

* Note: These scores may change depending upon the application and intuition.

Level	Description	Meaning	Example
C1	Easily controllable	Driver can react safely	Height change is slow, driver adjusts
C2	Normally controllable	Average driver can manage	Noticeable tilt, needs careful steering
C3	Difficult or uncontrollable	Driver cannot avoid accident	Sudden suspension collapse at speed

Table 1.1.3: C - Likelihood that the driver can avoid harm after the event starts.

Event ID	S*	E*	C*	ASIL*
HE1	3	3	2	C
HE2	2	2	2	B
HE3	3	2	2	B
HE4	3	2	1	B
HE5	2	1	3	QM (Quality Management)

Table 1.1.4: ASIL Score is then determined by ASIL Determination Matrix.

1.1.5 Derive Mechatronics Safety Goals.

Solution:

Safety Goal ID	Description	Related Hazard(s)	ASIL
SG1	Prevent uncontrolled or asymmetric height changes during driving.	H1, H3, H4	C
SG2	Detect and isolate faulty height sensors to prevent wrong adjustments.	H2	B
SG3	Maintain stable vehicle height even under partial system failure.	H1, H5	B
SG4	Ensure safe stop and stable height on power loss.	H5	QM

Task 1.2: Measurement Error and Accuracy

A pressure gauge with a measurement range of 0 - 10 bar has a quoted inaccuracy of 1.0% of the full-scale reading.

1.2.1 What is the maximum measurement error expected for this instrument?

Solution:

The maximum error expected in any measurement reading is 1.0% of the full-scale reading, which is 10 bar for this particular instrument. Hence, the maximum likely error is:

$$\text{Maximum error} = 1.0\% \times 10 \text{ bar} = 0.1 \text{ bar}$$

1.2.2 What is the likely measurement error expressed as a percentage of the output reading if this pressure gauge is measuring a pressure of 1 bar?

Solution:

The maximum measurement error is a constant value related to the full-scale reading of the instrument, irrespective of the magnitude of the measured quantity.

Actual Pressure	Max Error (bar)	Error as % of Reading
10 bar (full scale)	0.1	1.0%
5 bar	0.1	2.0%
1 bar	0.1	10.0%
0.5 bar	0.1	20.0%

Table 1.2.1: Relative measurement errors at different pressures

At lower readings, the relative error becomes much larger. Therefore, if we are measuring pressures between 0 and 1 bar, we would not use an instrument with a measurement range of 0 - 10 bar.

Task 1.3: Measurement Precision

The width of a room is measured 10 times by an ultrasonic sensor and the following measurements are obtained (in meters):

3.521, 3.539, 3.538, 3.536, 3.525, 3.522, 3.534, 3.523, 3.527, 3.529

The width of the same room is then measured by a calibrated steel tape that gives a reading of 3.520 m, which can be taken as the correct value for the width of the room.

1.3.1 What is the measurement precision of the ultrasonic sensor?

Solution:

The mean (average) value of the 10 measurements made with the ultrasonic sensor is:

$$\bar{x} = 3.529 \text{ m}$$

The maximum deviation below this mean value is -0.008 m and the maximum deviation above the mean value is 0.01 m. Thus, the precision of the ultrasonic rule can be expressed as $\max(-0.008, 0.01) = 0.01 \text{ m}$.

$$\text{Precision} = 0.01 \text{ m}$$

1.3.2 What is the maximum measurement inaccuracy of the ultrasonic sensor?

Solution:

The correct value of the room width has been measured as 3.520 by the calibrated steel rule. All ultrasonic sensor measurements are above this, with the largest value being 3.539 m. The maximum

measurement error can be calculated as:

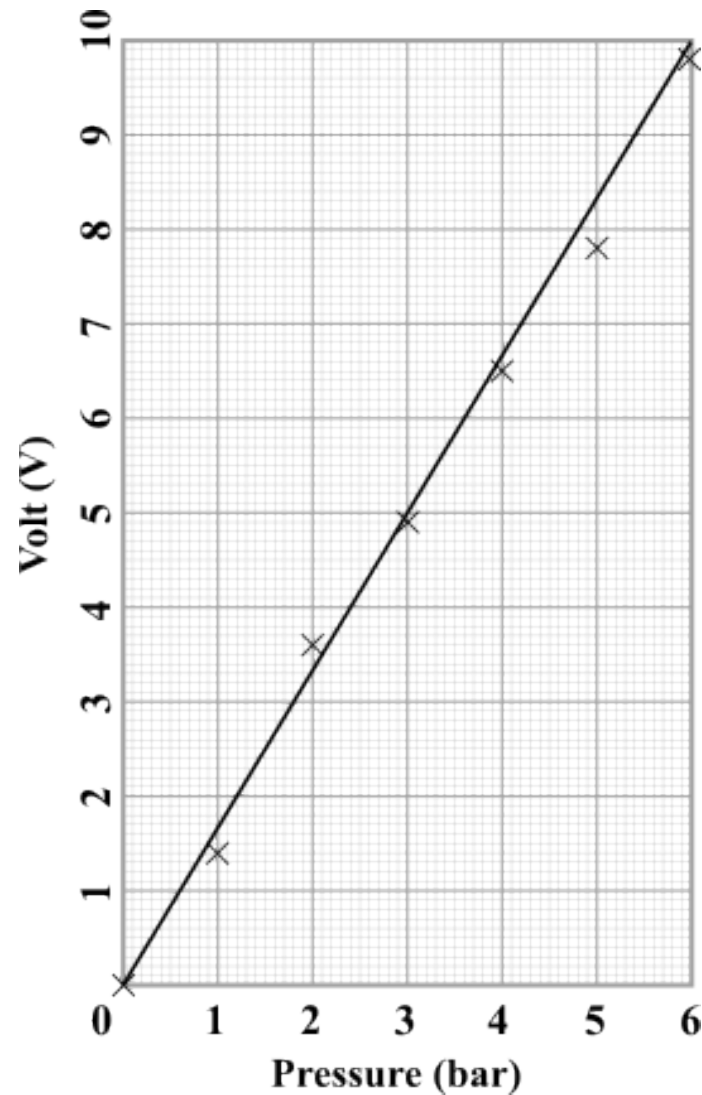
$$\text{Maximum error} = 3.539 - 3.520 = 0.019 \text{ m (19 mm)}$$

Thus, the maximum measurement inaccuracy can be expressed as:

$$\text{Inaccuracy} = 19 \text{ mm}$$

Task 1.4: Linearity

Suppose the instrument characteristic shown in figure below is that of a pressure sensor, in which the input units are pressure in bars from 1 to 6 bar, and the output units are expressed in volts from 1 to 10 V.



1.4.1 What is the maximum nonlinearity expressed as a percentage of the full-scale deflection?

[Solution:](#)

The maximum nonlinearity is the maximum deviation of any data point on the figure away from a straight line drawn through the data points. This deviation is indicated by the thick vertical line in the figure. The length of this line is 0.5 V, while the full-scale deflection (FSD) is 10.0 V.

Thus, the maximum nonlinearity is calculated as:

$$\text{Nonlinearity} = \frac{0.5}{10} \times 100 = 5\%. \quad (1.4.1)$$

Therefore, the maximum nonlinearity is 5 % of the full-scale deflection.

1.4.2 What is the resolution of the sensor as determined by the instrument characteristic given?

Solution:

The resolution of the sensor, as determined from the graph, is the smallest change in input pressure that can be detected. For the given graph, the naked eye cannot reliably distinguish any smaller change than one small square on the graph paper, corresponding to one-tenth of a unit (0.1 bar).

Hence, the resolution of the pressure sensor is:

$$\text{Resolution} = 0.1 \text{ bar}. \quad (1.4.2)$$

Task 1.5: Sensitivity

The following resistance values of a platinum resistance thermometer were measured at a range of temperatures. Determine the measurement sensitivity of the instrument in ohms per degree Celsius ($\frac{\Omega}{^\circ\text{C}}$).

Table 1.5.1: Measured resistance values of a platinum resistance thermometer.

Resistance (Ω)	Temperature ($^\circ\text{C}$)
307	200
314	230
321	260
328	290

1.5.1 Determine the measurement sensitivity of the instrument.

Solution:

From the data in ??, it can be observed that the relationship between resistance and temperature is approximately linear. For a temperature change of $\Delta T = 30^\circ\text{C}$, the corresponding change in resistance is $\Delta R = 7\Omega$.

Hence, the measurement sensitivity of the instrument is:

$$S = \frac{\Delta R}{\Delta T} = \frac{7}{30} = 0.233 \frac{\Omega}{^\circ\text{C}}. \quad (1.5.1)$$

Therefore, the sensitivity of the platinum resistance thermometer is $0.233 \frac{\Omega}{^{\circ}\text{C}}$.

Task 1.6: Zero Drift

This table shows the output measurements of a voltmeter under two sets of conditions:

1. use in an environment kept at 20°C , which is the temperature at which it was calibrated; and
2. use in an environment at 50°C .

Table 1.6.1: Voltmeter readings at different ambient temperatures.

Input	Reading at 20°C	Reading at 50°C
1	10.2	10.5
2	20.3	20.6
3	30.7	31.0
4	40.8	41.1

1.6.1 Determine the zero drift and the zero drift coefficient.

Solution:

From ??, it is observed that each corresponding reading at 50°C is consistently 0.3 V higher than the reading at 20°C .

Hence, the **zero drift** is:

$$\Delta V = 0.3\text{ V}. \quad (1.6.1)$$

The **zero drift coefficient** is the drift magnitude divided by the temperature change that caused it:

$$k = \frac{\Delta V}{\Delta T} = \frac{0.3}{50 - 20} = \frac{0.3}{30} = 0.01\text{ V}/^{\circ}\text{C}. \quad (1.6.2)$$

Therefore, the zero drift coefficient of the voltmeter is $0.01\text{ V}/^{\circ}\text{C}$.